

SOY: MYTHS AND TRUTHS

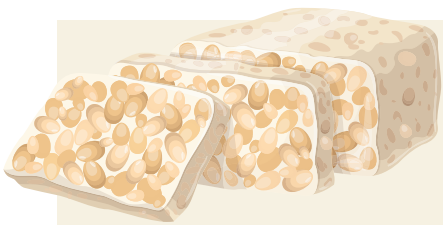
SOY & BREAST CANCER:

MYTH

- Soy increases the risk of breast cancer
- Soy acts like human estrogen
- Breast cancer survivors should avoid soy
- Soy supplements are the same as soy foods

FACT

- Human studies show soy is safe even for estrogen receptor (ER)–positive breast cancers.
- Women who eat moderate amounts of soy often have lower risk of recurrence and better survival.
- Soy isoflavones often block stronger estrogen from attaching to cells, acting more like a buffer than a trigger.
- Research shows survivors who eat soy foods often live longer and have lower recurrence rates.
- Supplements are not recommended. Whole soy foods like tofu, tempeh, soy milk, and edamame are safe and provide many nutrients.



SOY & MENOPAUSE:

MYTH

- Soy works like hormone therapy for hot flashes.

FACT

Evidence is mixed. Isoflavones are not a substitute for estrogen therapy, though some women may still notice benefits after regular use for 3+ months.

Notes:

- Isoflavones stay in the blood for only 6–8 hours, so soy should be eaten more than once a day.
- Some women benefit more if their gut bacteria can make equol from isoflavones.
- The North American Menopause Society does not recommend soy supplements due to inconsistent evidence, but soy foods are safe.

SOY & MALE FEMINIZATION:

MYTH

- Soy causes breast growth in men.

FACT

- A large review of 41 clinical trials found no effect of soy on male hormones.
- Two placebo-controlled trials found no evidence soy causes gynecomastia.
- Soy does not affect fertility or hormone levels in men or children.



SOY & THYROID:

MYTH

- Soy harms thyroid function.

FACT

Isoflavones can theoretically affect enzymes in thyroid hormone production, but **clinical studies show no harm.**

- A 2019 meta-analysis of 18 trials found no effect of soy or isoflavones on thyroid hormone levels.
- Concerns about soy with low iodine intake (2012) were not supported in later research.
- A 2011 study suggested soy might affect subclinical hypothyroidism, but repeating with higher doses found no adverse effect.
- No clinical trials to date on soy in Hashimoto's or Graves' disease.

SOY & SOY ISOLATES:

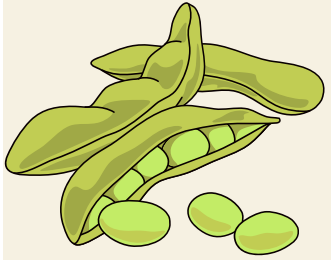
MYTH

- Foods with added soy protein or fiber are higher in isoflavones than whole soy.

FACT

- Soy protein isolates contain only about 10–20% of the isoflavone content found in whole soy.
- Whole soy foods provide approximately 3–4 mg of isoflavones per gram of protein (≈25 mg per serving).
- Isolates: 0.25–1 mg per gram protein.

Next >>>



SOY & ANTI-NUTRIENTS:

MYTH

- Soy's phytates and lectins make it unhealthy by blocking minerals and harming digestion.

FACT

- Soaking, cooking, and fermenting soy reduce these compounds.
- Fermented soy foods like tempeh, miso, and natto have very low levels.
- In a varied whole-food diet, they're not a problem.

SOY & EVERYDAY CONSUMPTION:

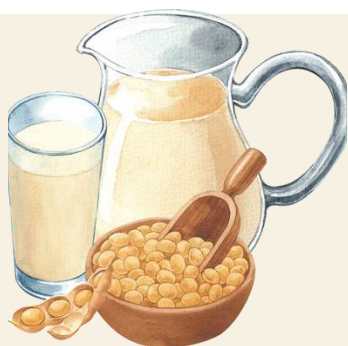
MYTH

- Eating a lot of soy is unsafe.
- Asians mostly eat fermented soy.

FACT

- In Asia, people average 35–40 mg of isoflavones daily, and some consume up to 100 mg (~4 servings) with no safety concerns.
- Most soy intake is unfermented—over 95% in China, ~70% in Korea, and about 50% in Japan.

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SOY & PRODUCTION:

MYTH

- Most soy foods people eat are GMO.
- GMO soy is unsafe to eat.

FACT

- Over 90% of GMO soy is fed to livestock, not humans.
- Tofu, soy milk, and edamame in stores are easy to find as non-GMO (organic).
- GMO soy has been studied extensively and is considered safe. However, consume organic (non-GMO) if you are still concerned.

SOY & EVERYDAY CONSUMPTION:

MYTH

- Harvesting soy is bad for the planet.

FACT

- The majority of soy land use is for animal feed.
- Eating soy directly—like tofu, tempeh, and edamame—uses far fewer resources and has a much lighter environmental footprint.

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RECOMMENDATIONS:

THYROID:

For people on thyroid medication:

- Soy (like calcium or iron) can reduce medication absorption.
- Take thyroid meds on an empty stomach, wait 1–3 hours before soy or supplements.
- If you eat soy consistently at the same times, your doctor can adjust your dose, so you may not need to wait each time.

SOY ALLERGY:

- Soy allergy is uncommon, affecting about 0.3–0.4% of the population. As in adults, soy allergy prevalence among children is the lowest among the Big 9.
- For those with a true soy allergy, avoiding soy is necessary, but for everyone else, soy foods are safe and beneficial.

HOW MUCH SOY:

- No established upper limit for soy foods.
- Up to 100 mg of soy isoflavones per day (about 4 servings) has been shown to be safe.



BOTTOM LINE

Isoflavones are not estrogens—they act differently in the body.

Soy does not raise hormone levels in men or women.

Whole soy foods are safe and may even protect against cancer recurrence and lower cholesterol.



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